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### Seal assembly, and system

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#### Abstract

A seal assembly, including a seal including a first lattice impregnated with a conformable resilient material, a backup including a second lattice impregnated with a conformable resilient material, the second lattice having a greater rigidity than the first lattice. A borehole system including a borehole in a subsurface formation, a string in the borehole, and a seal assembly disposed within or as a part of the string.

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## **Background/Summary**

### **BACKGROUND**

(1) In the resource recovery and fluid sequestration industries, sealing various parts relative to other parts is ubiquitous. Challenges abound due to temperature ranges and fluctuations, pressure ranges and fluctuations, and the complexities of systems to manage these. The art is always receptive to developments that improve the state of the art.

### **SUMMARY**

(2) An embodiment of a seal assembly, including a seal including a first lattice impregnated with a conformable resilient material, a backup including a second lattice impregnated with a conformable resilient material, the second lattice having a greater rigidity than the first lattice.

(3) An embodiment of a borehole system including a borehole in a subsurface formation, a string in the borehole, and a seal assembly disposed within or as a part of the string.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a schematic cross section view of a seal assembly as disclosed herein;

(3) FIG. 2 is a schematic cross section view of another embodiment of a seal assembly as disclosed herein;

(4) FIGS. 3 and 4 are schematic illustrations of the seal assembly of FIG. 1 in run in and set positions;

(5) FIGS. 5 and 6 are schematic illustrations of an alternate configuration of seal assembly in run in and set positions;

(6) FIGS. 7 and 8 are schematic illustrations of another alternate configuration of seal assembly in run in and set positions; and

(7) FIG. 9 is a view of a borehole system including a seal assembly as disclosed herein.

### **DETAILED DESCRIPTION**

(8) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(9) Referring to FIG. 1, a first embodiment of a seal assembly 10 is illustrated. Seal assembly 10 comprises a seal 12 including a first lattice 14 impregnated with a conformable resilient material 16. The lattice 14 is a geometric structure that exhibits a rigidity that is higher than that of the conformable resilient material 16. Lattice 16 may be of a metal such as stainless steel or a plastic such as polyether ether ketone (PEEK) and may have geometric shapes such as cubic, honeycomb, or other geometries. The conformable resilient material is a softer material that is suitable for creating a seal when urged into contact with another structure such as a casing. In embodiments, the material 16 may be a rubbery material such as Polytetrafluoroethylene (PTFE), Ethylene Propylene Diene Monomer (EPDM), Perfluoroalkoxy alkane (PFA), and other similar materials. In

each case the material **16** is injected into the lattice **14** or is overmolded on the lattice **16**.

(10) Still referring to FIG. 1, a backup **18** includes a second lattice **20** impregnated with a conformable resilient material **22** (which may be the same as material **16** or may be other similar materials. The second lattice **20** may be of the same rigidity as the first lattice **14** or may be of a higher rigidity than the first lattice **14**. If the lattice **20** is the same rigidity as the lattice **14**, as is illustrated in FIG. 1, then a particulate additive **24** is added to increase overall rigidity (lattice **20**, material **22**, and particulate **24**) of the backup **18** relative to the overall rigidity (lattice **14**, and material **16**) of the seal **12**. Alternatively, the second lattice **20** may have a greater rigidity than the first lattice **14** and be used without any particulate additive **24** as is illustrated in FIG. 2, where the size of the internal repeating shapes defined by the lattice **14** is different from the size of the shapes defined by the lattice **20**. In embodiments, the greater rigidity may be due to an increase in the thickness of material used for the lattice **20** or may be due to the dimensions of the repeating units of the lattice in the lattice **20** vs lattice **14**. In either case, the overall rigidity of the seal **12** and the backup **18** will be different in that the backup **18** will be of a higher rigidity than the seal **12**. In embodiments where a particulate additive **24** is employed, the additive **24** may be spherical objects of stainless steel or other metals or materials not likely to plastically deform under loads associated with setting of the seal assembly **10**. “Roundy” geometries improve flow characteristics of the additive **24** in the material **22**. By “roundy” it is meant, spherical, oblate spheroids, prolate spheroids, tori, and other geometric shapes that have been rounded where corners, vertexes, or edges are normally presented. The additive **24** significantly increases rigidity of the backup **18**. In an embodiment, a ratio of rigidity of backup to seal is about 1.1:1 to about 2:1, achievable with lattice **14** to lattice **20** difference alone, lattice **14** to lattice **20** difference plus additive **24**, or lattice **14** and lattice **20** being the same but with an additive **24**. In another embodiment, a volume percentage of lattice **20** and/or additive **24** to the total volume of the backup **18** is about 50% to about 85%. The volume percentage of lattice **14** in the seal **12** will be less than the percentage in the backup **18**.

(11) It is to be understood that in at least some of the embodiments, the seal **12** and backup **18** may be a single unit or may be different components like rings that are stacked next to one another. Further, the lattice **14** and **20** may be contiguous, with and different conformable resilient materials **16** and **22** impregnated in the locations illustrated for the seal **12** and backup **18**.

(12) Referring to FIGS. 3 and 4, the embodiment of FIG. 1 is illustrated in situ upon a mandrel **30** and inside of a casing **32** (any tubular structure). Gauge rings **34** and **36** are also illustrated and are clearly not compressing the seal assembly **10** in FIG. 3. In FIG. 4, however, the gauge rings **34** and **36** have been moved closer together to squeeze the seal assembly **10**. The less rigid seal **12** expands radially outwardly more quickly than the more rigid backups **18** forming a seal against the casing **32**. The backups **18** also expand radially outwardly into contact with the casing **32** and due to their rigidity prevent extrusion of the seal **12** during use.

(13) Referring to FIGS. 5 and 6, an alternate embodiment is illustrated that uses different shapes for the seal **12** and backups **18** to promote certain shapes of radial expansion upon setting, as illustrated. It will be noted that the shape of the gauge rings is also different in that they are angled toward each other as then increase in distance from the mandrel **30**. The embodiment increases extraction resistance by preferentially causing the backups **18** to angle toward one another.

(14) Referring to FIGS. 7 and 8, yet another embodiment is illustrated having anchor flanges **38**. This embodiment exhibits a complex shape for the seal **12** and the backups **18**. The anchor flanges **38** tend to anchor the backups under the seal **12** during pressure reversals.

(15) Referring to FIG. 9, a borehole system **40** is illustrated. The system **40** comprises a borehole **42** in a subsurface formation **44**. A string **46** is disposed within the borehole **42**. A seal assembly **10** as disclosed herein is disposed within or as a part of the string **46**.

(16) Set forth below are some embodiments of the foregoing disclosure:

(17) Embodiment 1: A seal assembly, including a seal including a first lattice impregnated with a

conformable resilient material, a backup including a second lattice impregnated with a conformable resilient material, the second lattice having a greater rigidity than the first lattice.

(18) Embodiment 2: The seal assembly as in any prior embodiment, wherein the backup has a rigidity in a ratio of about 1.1:1 to about 2:1 with the seal.

(19) Embodiment 3: The seal assembly as in any prior embodiment, further including a particulate additive dispersed in the conformable resilient material in the second lattice.

(20) Embodiment 4: The seal assembly as in any prior embodiment, wherein the particulate additive is a spherical additive.

(21) Embodiment 5: The seal assembly as in any prior embodiment, wherein the additive is stainless steel.

(22) Embodiment 6: The seal assembly as in any prior embodiment, wherein the additive is of curvilinear geometry.

(23) Embodiment 7: The seal assembly as in any prior embodiment, wherein the seal and the backup form a single piece structure.

(24) Embodiment 8: The seal assembly as in any prior embodiment, wherein the first lattice and/or the second lattice defines shapes that are cubic in geometry.

(25) Embodiment 9: The seal assembly as in any prior embodiment, wherein first lattice defines internal repeating shapes; the second lattice defines internal repeating shapes, and the shapes of the first lattice are smaller in dimension than the shapes of the second lattice.

(26) Embodiment 10: The seal assembly as in any prior embodiment, wherein at least one of the first and second lattices comprise a metal material.

(27) Embodiment 11: The seal assembly as in any prior embodiment, wherein the at least one of the first and second lattices comprise a plastic material.

(28) Embodiment 12: The seal assembly as in any prior embodiment, wherein the metal material is stainless steel.

(29) Embodiment 13: The seal assembly as in any prior embodiment, wherein the conformable resilient material of the seal material and the conformable resilient material of the backup material are the different materials.

(30) Embodiment 14: The seal assembly as in any prior embodiment, wherein the conformable resilient material of the seal material and the conformable resilient material of the backup material are the same material.

(31) Embodiment 15: The seal assembly as in any prior embodiment, wherein the conformable resilient material of the seal material and/or the conformable resilient material of the backup material comprises a polymeric material.

(32) Embodiment 16: The seal assembly as in any prior embodiment, wherein the polymeric material includes poly tetra fluoroethelene, ethylene propylene diene monomer, perfluoroalkoxy alkane, and combinations including at least one of the foregoing.

(33) Embodiment 17: The seal assembly as in any prior embodiment, wherein the seal exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, that is a parallelogram.

(34) Embodiment 18: The seal assembly as in any prior embodiment, wherein the seal exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, having a larger dimension radially inwardly and a smaller cross section radially outwardly,

(35) Embodiment 19: The seal assembly as in any prior embodiment, wherein the backup exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, that is a parallelogram.

(36) Embodiment 20: A borehole system including a borehole in a subsurface formation, a string in the borehole, and a seal assembly as in any prior embodiment disposed within or as a part of the string.

(37) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the invention (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context.

Further, it should be noted that the terms “first,” “second,” and the like herein do not denote any order, quantity, or importance, but rather are used to distinguish one element from another. The terms “about,” “substantially” and “generally” are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” and/or “substantially” and/or “generally” can include a range of +8% a given value.

(38) The teachings of the present disclosure may be used in a variety of well operations. These operations may involve using one or more treatment agents to treat a formation, the fluids resident in a formation, a borehole, and/or equipment in the borehole, such as production tubing. The treatment agents may be in the form of liquids, gases, solids, semi-solids, and mixtures thereof. Illustrative treatment agents include, but are not limited to, fracturing fluids, acids, steam, water, brine, anti-corrosion agents, cement, permeability modifiers, drilling muds, emulsifiers, demulsifiers, tracers, flow improvers etc. Illustrative well operations include, but are not limited to, hydraulic fracturing, stimulation, tracer injection, cleaning, acidizing, steam injection, water flooding, cementing, etc.

(39) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

## Claims

1. A borehole seal assembly configured for deployment in a subsurface formation having a temperature range higher than ambient surface temperature, comprising: a seal including a first lattice impregnated with a conformable resilient material; a backup including a second lattice impregnated with a conformable resilient material, the second lattice having a greater rigidity than the first lattice, further including a particulate additive dispersed in the conformable resilient material in the second lattice, the particulate additive being of roundy geometry and being configured to not plastically deform under loads associated with setting of the seal assembly in the borehole, the particulate material storing energy when deformed, and wherein a volume percentage of the second lattice and/or additive to the total volume of the backup is about 50% to about 85%.
2. The seal assembly as claimed in claim 1, wherein the backup has a rigidity in a ratio of about 1.1:1 to about 2:1 with the seal.
3. The seal assembly as claimed in claim 2, wherein the particulate additive is a spherical additive.
4. The seal assembly as claimed in claim 2, wherein the additive is stainless steel.
5. The seal assembly as claimed in claim 1, wherein the seal and the backup form a single piece structure.
6. The seal assembly as claimed in claim 1, wherein the first lattice and/or the second lattice defines shapes that are cubic in geometry.
7. The seal assembly as claimed in claim 1, wherein first lattice defines internal repeating shapes; the second lattice defines internal repeating shapes, and the shapes of the first lattice are smaller in dimension than the shapes of the second lattice.
8. The seal assembly as claimed in claim 1, wherein at least one of the first and second lattices

comprise a metal material.

9. The seal assembly as claimed in claim 1, wherein the at least one of the first and second lattices comprise a plastic material.

10. The seal assembly as claimed in claim 8, wherein the metal material is stainless steel.

11. The seal assembly as claimed in claim 1, wherein the conformable resilient material of the seal material and the conformable resilient material of the backup material are the different materials.

12. The seal assembly as claimed in claim 1, wherein the conformable resilient material of the seal material and the conformable resilient material of the backup material are the same material.

13. The seal assembly as claimed in claim 1, wherein the conformable resilient material of the seal material and/or the conformable resilient material of the backup material comprises a polymeric material.

14. The seal assembly as claimed in claim 12, wherein the polymeric material includes poly tetra fluoroethelene, ethylene propylene diene monomer, perfluoroalkoxy alkane, and combinations including at least one of the foregoing.

15. The seal assembly as claimed in claim 1, wherein the seal exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, that is a parallelogram.

16. The seal assembly as claimed in claim 1, wherein the seal exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, having a larger dimension radially inwardly and a smaller cross section radially outwardly.

17. The seal assembly as claimed in claim 1, wherein the backup exhibits a cross section, taken in parallel to a longitudinal axis of the seal assembly, that is a parallelogram.

18. A borehole system comprising: a borehole in a subsurface formation; a string in the borehole; and a seal assembly as claimed in claim 1 disposed within or as a part of the string.

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